

Qt Programming for HTTP REST Clients

This article discusses Qt support for connectivity to IoT platforms like ThingSpeak using HTTP REST APIs. It also focuses on handling JSON data in terms of encoding, parsing, URL encoding, forming query strings, etc.



Qt is a cross-platform development framework designed to provide eye candy GUI features and to give rich API support for Web communication, graph plotting, data exchange, 3D visualisation, multimedia handling, location tracking, sensor interfacing, etc. Qt applications can be built for various desktop platforms like Linux, Mac OS and Windows; for embedded platforms like Raspberry Pi and QNX, and also for mobile platforms like Android, iOS and Windows Mobile.

This article explores Qt support for connectivity to IoT platforms like ThingSpeak using HTTP REST APIs, and also focuses on handling JSON data in terms of encoding, parsing, URL encoding, forming query strings, etc. The assumption is that readers have basic familiarity with Qt and RESTful operations using HTTP support. If you are new to the Qt environment, do refer to some of the previous articles published in earlier editions of *OSFY*, as listed in the *References*, before proceeding.

About HTTP REST and ThingSpeak

ThingSpeak is an open source platform for building IoT applications, which supports connectivity using HTTP REST APIs. One can send data using the POST method

with a request payload or the GET method with a query string, and retrieve data in JSON or XML formats using the GET method. Recently, support for sending data using MQTT Publish has also been added. ThingSpeak has good integration support for MATLAB from Mathworks, which helps it to offer better analysis and visualisation of data. It provides SDKs in various languages for connectivity or one can try to connect using any HTTP client of the preferred language.

In ThingSpeak, data is stored in terms of channels, and each channel is associated with an ID, name, description and various applicable fields along with some tags, geo-location, etc. For simple and better authentication, API keys are provided for read/write operations and a channel key for meta operations. The API keys of a specific channel can be embedded in a query string or post data for authentication. Secure HTTP (https) is preferred to prevent unauthorised access of API keys by eavesdroppers. A channel can also be made public during the creation or by changing the settings later, in which case no read key is required for viewing the data.

One can log in to ThingSpeak using a Mathworks account, which is free of cost, and create any number of channels